

An Adjoint Gabor System Which Is Not an R-Dual of Type I

Candidate counterexample for arXiv:1509.06263
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Status. Candidate counterexample, likely valid, pending specialist human review. For every integer oversampling factor $q \geq 2$, the construction gives a Gabor frame whose normalized adjoint system is not an R-dual of type I.

1 Source question

Stoeva and Christensen ask whether the normalized adjoint system of every Gabor frame is an R-dual of type I [1, p. 6, after Theorem 1.4]. With

$$(T_y f)(x) = f(x - y), \quad (E_\xi f)(x) = e^{2\pi i \xi x} f(x),$$

their question is whether, whenever $\{E_{mb}T_{na}g\}_{m,n \in \mathbb{Z}}$ is a frame for $L^2(\mathbb{R})$, the Riesz sequence

$$\left\{ \frac{1}{\sqrt{ab}} E_{m/a} T_{n/b} g \right\}_{m,n \in \mathbb{Z}}$$

must be an R-dual of type I. The source records positive answers for tight Gabor frames and Gabor Riesz bases, but leaves the general case open.

The source also gives the criterion needed below. If S is the frame operator of a frame and S_Ω is the frame operator of a proposed Riesz sequence on its closed span, then type-I R-duality requires an antiunitary map Λ satisfying

$$S_\Omega = \Lambda S \Lambda^{-1}. \tag{1}$$

Indeed, this is the necessary direction of Lemma 1.3(iii) in [1].

2 Proof intuition

At integer oversampling, there is a Parseval Gabor window whose normalized adjoint system occupies only one of q Zak components. The adjoint shifts act as scalar coordinate multipliers on all q components. We deform the Parseval window by a positive invertible multiplier ϕ which has simple spectrum on the two-dimensional Zak base.

The deformation makes the primal frame operator the q -fold amplification $M_\phi \otimes I_q$, while the adjoint frame operator is the one-copy operator M_ϕ . Choosing ϕ essentially one-to-one makes the second operator cyclic. The first operator has no cyclic vector when $q \geq 2$. Since antiunitary equivalence preserves cyclicity, the necessary condition (1) fails.

is a Riesz basis for $L^2(\mathbb{R})$. Thus, the relation between the Gabor system $\{E_{mb}T_{na}g\}_{m,n \in \mathbb{Z}}$ and the system $\{\frac{1}{\sqrt{ab}}E_{m/a}T_{n/b}g\}_{m,n \in \mathbb{Z}}$ corresponds exactly to the relation between the sequence $\{f_i\}_{i \in I}$ and its R-duals, see Theorem [1.2](#). However, it is still not known whether this is a coincidence, or the R-duals of type I actually generalize the duality principle. That is, given a Gabor frame $\{E_{mb}T_{na}g\}_{m,n \in \mathbb{Z}}$ for $L^2(\mathbb{R})$, we do not know whether the Gabor system $\{\frac{1}{\sqrt{ab}}E_{m/a}T_{n/b}g\}_{m,n \in \mathbb{Z}}$ always can be realized as an R-dual of type I of $\{E_{mb}T_{na}g\}_{m,n \in \mathbb{Z}}$ (by [11](#), the answer is affirmative for tight Gabor frames and Gabor Riesz bases). This is the motivation for the introduction of the other types of R-duals in [13](#). In fact, in [13](#) it was shown that the R-duals of type III generalize the duality principle for all Gabor systems. However,

Figure 1: The open question in arXiv:1509.06263, PDF page 6.

3 The Zak model

Fix an integer $q \geq 2$ and put

$$a = 1, \quad b = \frac{1}{q}, \quad h = q^{-1/2} \mathbf{1}_{[0,1)}.$$

Lemma 1. *The Gabor system*

$$\{E_{m/q}T_n h\}_{m,n \in \mathbb{Z}}$$

is a Parseval frame for $L^2(\mathbb{R})$. Its normalized adjoint system is

$$\{\sqrt{q} E_m T_{qn} h\}_{m,n \in \mathbb{Z}} = \{E_m \mathbf{1}_{[qn, qn+1)}\}_{m,n \in \mathbb{Z}}, \quad (2)$$

which is an orthonormal sequence.

Proof. For each n , the atoms are supported on $[n, n+1)$. Write $m = qk + j$ with $0 \leq j < q$. For fixed j , the functions

$$e^{2\pi i(k+j/q)x} \mathbf{1}_{[n, n+1)}(x), \quad k \in \mathbb{Z},$$

form an orthonormal basis of $L^2([n, n+1))$. Multiplication by $q^{-1/2}$ and summation over the q residue classes gives Parseval's identity. Formula (2) follows from $ab = 1/q$; within each unit interval it is the Fourier orthonormal basis, and the intervals indexed by n are disjoint. $\square$

Let $X = [0, 1)^2$ with coordinates (t, θ) , and let $e_0, \dots, e_{q-1}$ be the standard basis of $\mathbb{C}^q$. Define the q -step Zak transform by

$$(\mathcal{Z}_q f)_r(t, \theta) = \sum_{k \in \mathbb{Z}} f(t + r + qk) e^{-2\pi i k \theta}, \quad 0 \leq r < q. \quad (3)$$

Partitioning $\mathbb{R}$ into the sets $t + r + qk$ and applying Fourier-series Parseval in k shows that

$$\mathcal{Z}_q : L^2(\mathbb{R}) \longrightarrow L^2(X; \mathbb{C}^q)$$

is unitary. Direct substitution gives

$$\mathcal{Z}_q E_1 \mathcal{Z}_q^{-1} = M_{e^{2\pi i t}} \otimes I_q, \quad \mathcal{Z}_q T_q \mathcal{Z}_q^{-1} = M_{e^{-2\pi i \theta}} \otimes I_q. \quad (4)$$

Moreover,

$$\mathcal{Z}_q h = q^{-1/2} e_0. \quad (5)$$

Under $\mathcal{Z}_q$, the atoms in (2) become

$$e^{2\pi i m t} e^{-2\pi i n \theta} e_0, \quad m, n \in \mathbb{Z}.$$

Consequently their closed span is

$$\mathcal{K} = L^2(X) e_0 \subset L^2(X; \mathbb{C}^q). \quad (6)$$

4 Construction of the window

We need a bounded scalar multiplier with simple spectrum. Give each $t, \theta \in [0, 1)$ its binary expansion which is not eventually all ones,

$$t = .t_1 t_2 t_3 \dots, \quad \theta = .s_1 s_2 s_3 \dots,$$

and interleave the digits:

$$\psi(t, \theta) = \sum_{j \geq 1} \left(t_j 2^{-(2j-1)} + s_j 2^{-2j} \right), \quad \phi = 1 + \psi. \quad (7)$$

Then $\phi : X \rightarrow [1, 2)$ is a Borel injection. Its inverse on its range is Borel: the odd and even binary digits recover t and θ .

On $L^2(\mathbb{R})$ define the positive invertible operator

$$A = \mathcal{Z}_q^{-1} (M_\phi \otimes I_q) \mathcal{Z}_q \quad (8)$$

and the window

$$g = A^{1/2} h. \quad (9)$$

Equivalently, this is the explicit Zak-coordinate prescription

$$\mathcal{Z}_q g = q^{-1/2} \sqrt{\phi} e_0. \quad (10)$$

The two unitaries in (4) generate all scalar multiplication operators on $L^2(X; \mathbb{C}^q)$ by their joint bounded Borel functional calculus. Thus A belongs to the von Neumann algebra generated by E_1 and T_q . These two generators commute with one another. They also commute with every primal-lattice shift $E_{m/q} T_n$, since

$$E_1 T_n = T_n E_1, \quad T_q E_{m/q} = E_{m/q} T_q \quad (m, n \in \mathbb{Z}).$$

It follows that A commutes with the entire primal lattice. Because the adjoint shifts are the products $E_m T_{qn}$, A commutes with them as well.

Theorem 1. *For every integer $q \geq 2$, the window g in (9) has the following properties:*

- (i) $\{E_{m/q} T_n g\}_{m, n \in \mathbb{Z}}$ is a Gabor frame for $L^2(\mathbb{R})$;

(ii) *its normalized adjoint system*

$$\{\sqrt{q} E_m T_{qn} g\}_{m,n \in \mathbb{Z}}$$

is not an R-dual of type I.

Thus the universal type-I R-duality question in [1] has a negative answer.

Proof. Let S_g be the frame operator of the primal system. Since $A^{1/2}$ commutes with every primal shift and the system generated by h is Parseval, Lemma 1 gives

$$S_g = A^{1/2} I A^{1/2} = A. \quad (11)$$

In particular the system generated by g is a frame, because $I \leq A \leq 2I$.

Write

$$\omega_{m,n}^h = \sqrt{q} E_m T_{qn} h, \quad \omega_{m,n}^g = \sqrt{q} E_m T_{qn} g.$$

Commutation with the adjoint shifts gives

$$\omega_{m,n}^g = A^{1/2} \omega_{m,n}^h. \quad (12)$$

The $\omega_{m,n}^h$ form an orthonormal basis of $\mathcal{K}$. Under $\mathcal{Z}_q$, $A^{1/2}$ is scalar multiplication by $\sqrt{\phi}$ and hence maps $\mathcal{K}$ bijectively onto itself. Therefore the closed span of the $\omega_{m,n}^g$ is again $\mathcal{K}$, and their frame operator on $\mathcal{K}$ is

$$S_\Omega = A^{1/2} P_{\mathcal{K}} A^{1/2}|_{\mathcal{K}} = A|_{\mathcal{K}}. \quad (13)$$

Using (8), (6), (11), and (13), the two operators whose antiunitary equivalence is required by (1) are thus

$$S_g \simeq M_\phi \otimes I_q \quad \text{on } L^2(X; \mathbb{C}^q), \quad S_\Omega \simeq M_\phi \quad \text{on } L^2(X). \quad (14)$$

We distinguish them by cyclicity. A vector u is cyclic for a self-adjoint operator B when the closed span of $\{F(B)u : F \text{ bounded Borel}\}$ is the whole Hilbert space. Since ϕ is a Borel injection with Borel inverse on its range, every bounded measurable function on X is a bounded Borel function of ϕ . Hence the constant function 1 is cyclic for M_ϕ on $L^2(X)$.

In contrast, $M_\phi \otimes I_q$ has no cyclic vector when $q \geq 2$. Indeed, given $v \in L^2(X; \mathbb{C}^q)$, if $v = 0$ there is nothing to prove; otherwise choose a nonzero measurable vector field $w \in L^2(X; \mathbb{C}^q)$ with $w(x) \perp v(x)$ almost everywhere. For example, using the first two coordinates one may rotate a nonzero coordinate pair pointwise, with a measurable partition according to the first nonzero coordinate. Then

$$\langle F(\phi)v, w \rangle = 0$$

for every bounded Borel F , so v is not cyclic.

Unitary or antiunitary equivalence of self-adjoint operators carries spectral projections to spectral projections and therefore preserves the existence of a cyclic vector. The two operators in (14) cannot be antiunitarily equivalent. The necessary type-I condition (1) fails, proving (ii). $\square$

Remark 1. *The frame is neither tight nor a Riesz basis. It is not tight because ϕ is nonconstant. The original Parseval frame is the union of q orthonormal bases on each translation interval and is redundant; applying the invertible operator $A^{1/2}$ preserves this redundancy. Thus the example lies genuinely outside the two positive classes recorded in the source.*

5 Verification, scope, and novelty

The proof is exact and has no computational dependency. The normalization of both Gabor systems, the two Zak-shift formulas, all commutation relations, the two frame-operator identities, and the cyclicity obstruction were checked separately in the companion `verification.md`. The construction uses only a bounded positive invertible operator and therefore produces a valid L^2 window.

The theorem answers the printed universal question in the negative. Its window is defined by a merely Borel Zak multiplier. It does not decide whether a type-I statement might remain true under additional regularity or time–frequency localization hypotheses on g .

Before promotion, the run’s registry, solution, attempt, and proof-gap indexes were searched using arXiv:1509.06263, the title, the authors, and combinations of *R-dual of type I*, *adjoint Gabor system*, *counterexample*, *Zak*, and spectral-multiplicity terms. Focused arXiv/web searches through 11 August 2026 found the source, its predecessor arXiv:1404.3183, and the 2024 paper [2], but no resolution or counterexample. The latter still describes the question as a famous problem and develops weak R-duality as a possible route.

Novelty confidence is moderate to high rather than conclusive: the bounded search found no prior answer, but a specialist search of the broader Gabor and noncommutative-torus literature is recommended. Human verification should focus on the joint Borel functional-calculus step and the application of Lemma 1.3(iii) of [1].

References

- [1] D. T. Stoeva and O. Christensen, *On various R-duals and the duality principle*, Operators and Matrices 10 (2016), no. 2, 371–385; arXiv:1509.06263. <https://arxiv.org/abs/1509.06263>.
- [2] H. Bansal, P. Devaraj, and S. Arati, *Characterizations of weak R-duality and its application to Gabor frames*, arXiv:2408.14952 (2024). <https://arxiv.org/abs/2408.14952>.